What Is Transportation Problem

Transportation Problem – Part 1: What Is It and What Is Its Purpose?

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1. Introduction

The Transportation Problem is a special type of Linear Programming (LP) problem. It focuses on finding the most cost-effective way to ship goods from multiple origins (suppliers) to multiple destinations (demand points). The goal is to satisfy all supply and demand constraints while minimizing total transportation cost.

Real-World Analogy

• Factories produce goods and ship them to warehouses.

• Warehouses then distribute products to retail stores.

• Retail stores fulfill customer orders.

In each step, the decision-maker must decide how many units to send from each origin to each destination, balancing costs and capacity.

2. Key Components

The transportation problem is built on four essential elements:

Component

Description

Example

Origins (Sources/Suppliers)

Places where goods are available. Each has a limited supply.

Factory A (supply = 100 units)

Destinations (Demand Points)

Places where goods are needed. Each has a specific demand.

Store X (demand = 80 units)

Unit Transportation Cost

Cost of shipping one unit from a specific origin to a specific destination.

$5 per unit from Factory A to Store X

Decision Variables

Amount to ship from each origin to each destination.

x_{ij} = units shipped from origin i to destination j

Notation

- Let there be m origins and n destinations.
- Supply at origin i: S_i
- Demand at destination j: D_j
- Cost per unit from origin i to destination j: c_{ij}
- Decision variable: x_{ij} (non-negative)

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3. Objective

The primary objective of the transportation problem is to:

Minimize total transportation cost = $(c_{ij} \times x_{ij})$ for all i and j

Subject to:

- Supply constraint: The total amount shipped from each origin cannot exceed its supply.

  $_j\, x_{ij} \le S_i$ for each origin i

- Demand constraint: The total amount received at each destination must meet its demand.

  $_i\, x_{ij} \ge D_j$ for each destination j

- Non-negativity: $x_{ij} \ge 0$ for all i, j

When total supply equals total demand, the problem is balanced. If not, dummy origins or destinations are added.

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4. Real-World Applications

The transportation problem is widely used across industries:

- Supply Chain Management – Optimizing shipments between factories, warehouses, and retailers.
- Logistics and Distribution – Planning delivery routes and load assignments.

• Emergency Resource Allocation – Distributing medical supplies, food, or water after disasters.
• Military Logistics – Moving troops, equipment, and supplies to bases.
• Telecommunication Network Routing – Assigning data traffic between servers and users.

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5. Historical Background

The transportation problem has a rich history in operations research:
• 1941 – Frank L. Hitchcock first formulated the problem in a paper titled "The Distribution of a Product from Several Sources to Numerous Localities."
• 1947 – Tjalling Koopmans independently developed similar ideas while studying economic allocation problems.
• 1951 – George Dantzig applied the simplex method to solve transportation problems, demonstrating its efficiency.
These contributions laid the foundation for modern network flow optimization.

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6. Why Is It Important?

The transportation problem is more than just a textbook exercise. It is a cornerstone of operations research for several reasons:
• Foundation for Network Flow Problems – It introduces concepts used in more complex models like transshipment, assignment, and minimum cost flow.
• Efficient Algorithms – Specialized methods (e.g., the transportation simplex, Vogel's approximation) solve it much faster than general LP solvers.
• Massive Cost Savings – Companies save millions of dollars annually by optimizing shipping routes and reducing fuel, labor, and inventory costs.
• Scalable – The model can handle hundreds of origins and destinations, making it practical for real-world logistics.

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